

Overview

Despite its central role in physics research and application, computation remains largely absent from the undergraduate physics curriculum. Physicists in academia, industry, and government routinely rely on computational tools for modeling, simulation, and data analysis, yet most physics majors graduate with little formal preparation in these skills. Multiple national reports have called for increased integration of computation in undergraduate physics education.

Integration of computation into the undergraduate physics curriculum is a department-wide institutional change process because it requires curricular-wide decisions and broad instructor engagement. Spreading computation integration over many departments requires application of a different change model, more suitable for a national change than for a single cohesive unit. Experts have proposed a *propagation model* with three phases: interactive development, interactive dissemination, and supporting adopters. We hypothesize that past efforts to promote computation in physics education have had limited impact on the discipline because they lacked structured frameworks for either institutional change or wide-scale propagation.

Intellectual Merit

We have formed a collaboration of nine physics departments spanning a wide array of institution types and missions. Each will apply Kotter's 8-step change model to integrate computation systematically and sustainably into their undergraduate physics curriculum. This work expands on a successful pilot at the University of Minnesota, supported by an IUSE-Institutional and Community Transformation Level 1 grant.

As part of adapting Kotter's model to conditions at local departments, collaboration members will develop and disseminate three products that would make it easier for other physics departments to embark on computation integration: (1) a consensus list of computation competencies and learning goals for physics undergraduates, with a curriculum map linking these goals to common courses. (2) a validated assessment instrument to measure both student learning outcomes and the effectiveness of change efforts; and (3) case studies documenting implementation approaches across a range of institutional contexts, along with a small set of computation integration change models that physics departments could adapt to guide their own efforts. To support broader adoption, we will carry out the interactive development and dissemination phases of the propagation model by working with an Ambassador cohort of 30 physics instructors. Fifteen Ambassadors, representing a wide array of institution types, including 2-year, 4-year, and PhD-granting institutions, have already been recruited.

The collaboration is co-led by the PI of the Minnesota pilot and a leader from the Partnership for Integrating Computation in Undergraduate Physics, a national community of practice. The team includes specialists in institutional change and assessment instruments and has a collective track record totaling more than 170 person-years of education research and instructional reform. The team will receive advice from an Advisory Board representing the perspectives of industry, government, national laboratory, artificial intelligence in physics education, and the American Association of Physics Teachers. An evaluator experienced with STEM projects will provide formative evaluations.

Broader Impacts

Integrating computation into undergraduate physics education will better prepare students for careers in research, industry, and education, strengthening the US scientific and technical workforce. It will give students the skills that are increasingly expected of them in professional settings. Embedding computational methods into the curriculum improves physics instruction because it supports more authentic engagement with current physics practices and gives a deeper conceptual understanding of the material by using perspectives only possible with computation. This initiative expands the ways students can see themselves participating in physics. Some students are drawn to the algorithmic, modeling, or coding aspects of the discipline. Including these elements in the curriculum makes physics more approachable and relevant to them. Finally, this project serves as a testbed for combining two complementary change models: Kotter's model for guiding department-level transformation, and the propagation model for scaling efforts across institutions. Lessons from the proposed work will inform future efforts to support systemic, long-lasting improvements in undergraduate STEM education.